

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

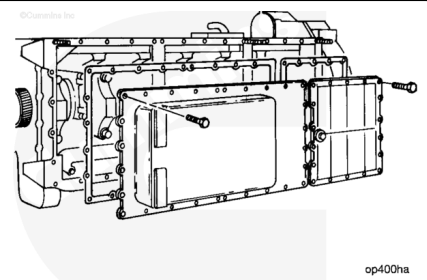
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

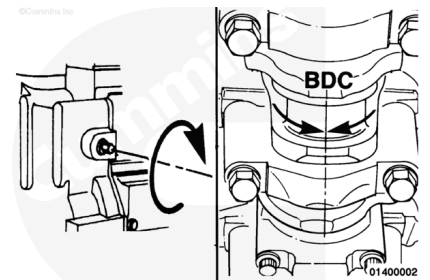


- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/20/20-013-009.html>)
- Drain the oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)
- Remove the filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html>)

- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html)
- Remove the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7.
(/qs3/pubsys2/xml/en/procedures/20/20-007-027-tr.html)
- Remove the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-035-tr.html)

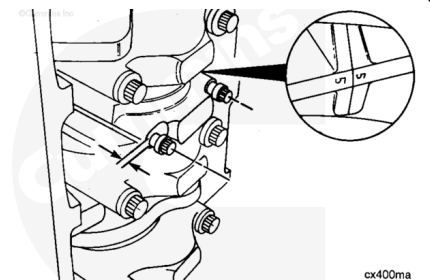
Remove

Use the barring mechanism to rotate the engine. Rotate the crankshaft to position a connecting rod at bottom dead center.



The connecting rods **must** have the cylinder number marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for correct markings. Use a steel stamp and mark any rod that is **not** correctly marked.

Loosen the capscrews until there is 6 mm [1/4 inch] of clearance between the rod cap and the capscrew head.

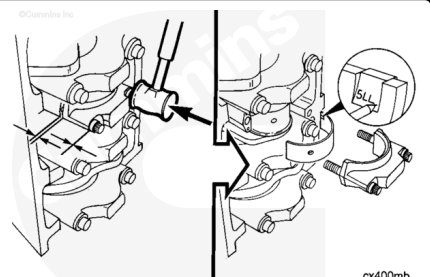


Use a mallet to tap the connecting rod capscrews until the connecting rod cap and rod separate.

Remove the capscrews and the cap.

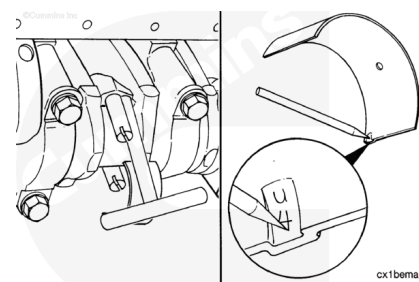
Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

Note : Mark the cylinder number and the letter "L" in the flat surface of the bearing tangs.



After installing two guide pins, Part Number 3375098, in the connecting rod, push the rod up far enough to allow the upper bearing shell to be removed. Use a “T handle” piston pusher to push the rod away from the crankshaft.

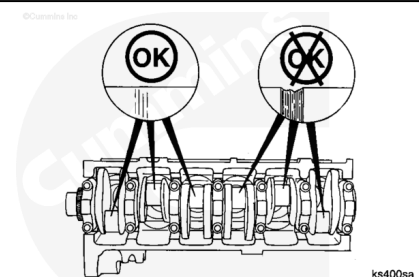
Remove the bearing shell, and mark the cylinder number and the letter “U” in the flat surface of the bearing tangs.



Clean and Inspect for Reuse

Use a clean, lint free paper towel to clean the rod bore, the cap, and the crankshaft rod journal.

Inspect the crankshaft rod journals for damage.

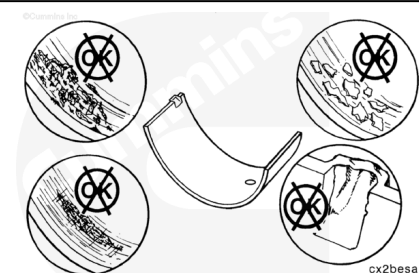


Clean the connecting rod bearing shells. Use a lint free cloth.

Inspect the bearing face on the front and back for:

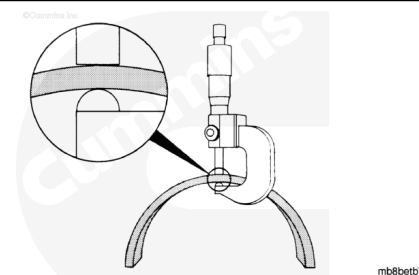
- Nicks
- Cracks
- Burrs
- Scratches
- Scrapes
- Gouges
- Embedded particles
- Fretting.

If any damage is found, the bearing **must** be replaced.



Use a ball-end micrometer to measure the thickness of the bearing in the wear area.

Use the following table for allowable dimensions of the various sizes of connecting rod bearings.



Connecting Rod Bearing Thickness

	mm		in
Standard	3.1250	MIN	0.1230
	3.1750	MAX	0.1250

0.010 Oversize	3.2520	MIN	0.1280
	3.3020	MAX	0.1300
0.020 Oversize	3.3790	MIN	0.1330
	3.4290	MAX	0.1350
0.030 Oversize	3.5060	MIN	0.1380
	3.5562	MAX	0.1400

If the bearings are **not** within specification, they **must** be replaced.

Install

⚠ CAUTION ⚠

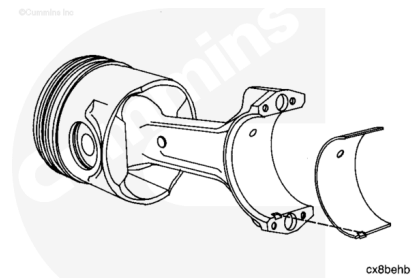
To reduce the possibility of engine damage, do not lubricate the back of the bearing shells.

⚠ CAUTION ⚠

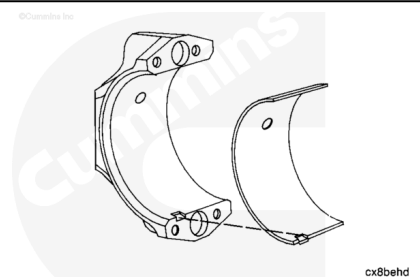
Do not reuse connecting rod bearings if either the connecting rod or crankshaft has been damaged.

Note : Used bearings must be installed in the same locations in which they were removed.

Lubricate the bearing shell with engine oil. Install the upper bearing shell in the connecting rod with the tang of the bearing in the slot of the connecting rod.

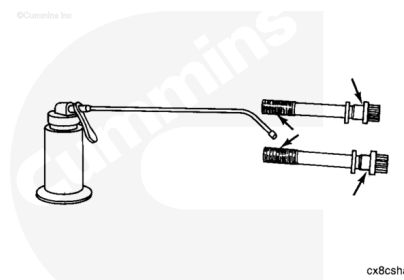


Install the lower bearing shell in the rod cap. Be sure the tang of the bearing shell is in the slot of the rod cap and the end of the bearing is even with the surface of the rod cap.



Use engine oil to lubricate the connecting rod capscrews and washers as illustrated.

Install the washers and capscrews in the connecting rod caps.



⚠ CAUTION ⚠

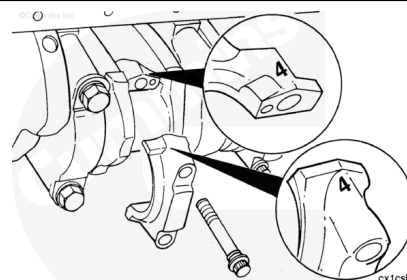
The connecting rods and caps are not interchangeable. The rods and the caps are machined as an assembly. Failure will result if the parts are mixed.

⚠ CAUTION ⚠

The cylinder number on the rod and cap must be the same. Failure will result if the parts are mixed.

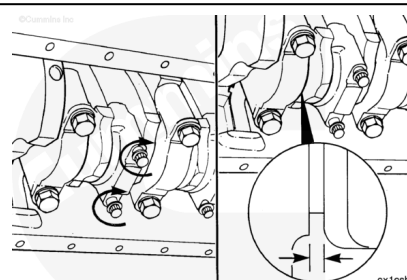
⚠ CAUTION ⚠

The side of the cap with the cylinder number marking (bearing tang side) must be toward the camshaft. Failure will result if the parts are not installed correctly.

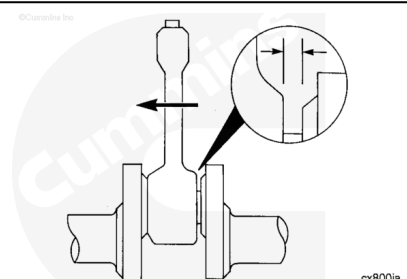


Install the connecting rod cap.

Tighten the capscrews alternately and evenly to pull the cap over the dowel pins. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-054-tr.html>)



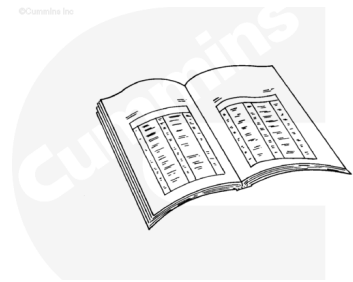
Check the side clearance between the rod and the crankshaft. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/20/20-001-054-tr.html>)



Finishing Steps

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- Install the suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-035-tr.html>)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html>)
- Install the lubricating oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-027-tr.html>)
- Install new oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-013-tr.html>)
- Fill the engine with oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)

- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/20/20-013-009.html>)
- Start the engine and check for leaks.

Last Modified: 20-Aug-2015
